



NEW BULGARIAN UNIVERSITY

DEPARTMENT OF COGNITIVE SCIENCE
AND PSYCHOLOGY



WHEN THERE IS NOTHING TO LOOK AT:

The role of visual spatial attention in
working memory

Borislav Parnarev

A thesis submitted in partial fulfillment of the requirements
for the degree of M.Sc. in Cognitive Science

Supervisor:
Assist. Prof. **Kiril Kostov**

Sofia, Bulgaria | 2021

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to everyone from the Cognitive Science and Psychology department for providing me with the building blocks that formed the foundation of this project. Your professionalism is deeply appreciated.

A big thank you to Kiril Kostov for his invaluable support and input in the development and completion of the current work.

I am grateful to Armina Janyan for her passionate enthusiasm in guiding us throughout the course of the program.

Thank you to Ivan Vankov for inspiring the project and to everyone else who took the time to review the current work and to provide their valuable thoughts and insights.

TABLE OF CONTENTS:

Introduction.....	5
Working memory, visual attention and the oculomotor system.....	7
Looking at nothing.....	12
Rationale of the current work.....	16
Experiment 1	
• Purpose	18
• Participants	18
• Apparatus and stimuli	18
• Procedure	19
• Design	20
• Results and discussion	21
Experiment 2	
• Purpose	25
• Participants	25
• Apparatus and stimuli	25
• Procedure	26
• Design	26
• Results and discussion	26
Experiment 3	
• Purpose	30
• Participants	30
• Apparatus and stimuli	30
• Procedure	31
• Design	31
• Results and discussion	31
Experiment 4	
• Purpose	35
• Participants	35
• Apparatus and stimuli	35
• Procedure	35
• Design	36
• Results and discussion	36
Additional Analysis	40
General discussion	45
References	50

ABSTRACT

It has been established that people sometimes tend to spontaneously look at empty locations, where relevant information has previously been presented. This phenomenon is referred to as looking at nothing and still echoes debates in the field of working memory, visual attention and eye movements. The earlier studies of the phenomenon (Spivey & Geng, 2001; Richardson & Spivey, 2000; Hoover & Richardson, 2008) revealed no functional role of these looks at blank locations in memory retrieval, supporting theories of embodiment and situated cognition. A different line of research (Ferreira et al., 2008; Johansson & Johansson, 2014; Scholz, Mehlhorm & Krems, 2016) proposed a model of integrated internal representations, postulating that the looking at nothing phenomenon is a mechanism that aids memory retrieval. The current work attempted to investigate these viewpoints, by conducting a series of experiments implementing reaction time to a probe dot. In four experiments participants were requested to keep four unique abstract shapes in their working memory. After a short retention interval, they were asked whether a cue stimulus was part of the initial memory array. While retrieving this information, on some trials a probe dot was presented on a location either congruent or incongruent to the cue stimulus's original location in the memory array. Stimulus onset asynchrony (SOA) was also manipulated in an attempt to establish the temporal dynamics of potential looks at blank locations. The results of the experiments obtained no consistent evidence of looking at nothing effects, preventing any discussion about the functionality of looks at blank locations in memory retrieval. Nevertheless, the current work discussed the lack of results both in terms of methodological issues, as well as the overall fragility of the effect.

INTRODUCTION

Close your eyes and imagine that you are at the doorstep of your kitchen. It is very likely that you will know the approximate location of your refrigerator, your coffee maker and most of the objects in the area. Even if you're kilometers away, most likely you are able to create a rather vivid representation of this very familiar environment. Now imagine that you're at the doorstep of the kitchen in a friend's house, that you have visited a while ago, only once. Are you still able to create the same detailed representation of the environment? The first example illustrates our ability to create and maintain detailed internal representations of the environment. We often rely on these representations to guide our behavior through our everyday activities. The second example emphasizes our dependency of the external world – the dynamic, rich source of information and its inevitable constraints.

Our environment is relatively steady and full of visual information, readily accessible to serve us as guideline or even a frame for achieving our goals. Objects usually don't vanish into thin air and we can most often direct our eyes to their location in attempt to receive detailed information about their features. Interestingly, sometimes even in the absence of the appropriate visual information, people tend to return their eyes to the last known location of an object that they require information about. This phenomenon is referred to as looking at nothing and is the focus of the current study.

The following chapters will examine the looking at nothing phenomenon and its underlying mechanisms in the context of research done in laboratory environment. Chapter 2 will briefly outline related studies of the fundamental mechanisms of visual attention, working memory and their influence on the oculomotor system. Chapter 3 will present some of the research done that focuses specifically on the looking at nothing phenomenon and the unsettled debates of its functionality.

The rationale of the current work will be issued in Chapter 4. From Chapter 5 onwards, the current experimental approach for studying the relationship between visual attention, working memory and the oculomotor system in the light of the looking at nothing phenomenon will be demonstrated. Chapter 9 will provide a general discussion of the results of the four experiments, the conclusions and possible future directions for investigating the phenomenon.

WORKING MEMORY, VISUAL ATTENTION AND THE OCULOMOTOR SYSTEM

Working memory as a term has been originally coined by Miller, Galanter and Pribram (1960), as a concept of a unitary short-term memory system. More than a decade later, the concept of working memory was adapted by Baddeley and Hitch (1974), as a title of their multiple-component model, proposing working memory as a central system, necessary for the performance of complex cognitive activities such as learning, comprehension and reasoning. According to their view, working memory is responsible for the storage of a limited amount of information, for a limited amount of time, allowing multimodal information to flow available, while performing single or multiple tasks.

Originally, the multiple-component model of working memory was established by three independent buffer systems. (1) a supervisory system - the Central Executive, that serves as a regulatory component in working memory, organizing and coordinating the operation of the following two subordinate systems, (2) a phonologically based store, involved in the processing of verbal and articulatory information, and (3) a visuospatial store, related to retaining visual and spatial information, that is no longer readily available in our environment. In their attempt to address certain issues, not captured by the original model, the authors proposed a fourth component- the episodic buffer (Baddeley & Logie, 1999). According to them, the episodic buffer is responsible for conscious awareness and some executive functions, by binding information from the subordinate systems, and from long-term memory, into a unified episodic representation.

The metaphor of a “spotlight” is the most common, and perhaps still the best way to present the concept of visual attention. Visual attention refers to the process of filtering the visual clutter and selecting bits of information, that are relevant for achieving short-term goals and support ongoing behavior (Posner, 1980). A more recent study, investigating visual attention, suggests that directing attention to information, available in our working memory is similar to directing attention to information available in our external environment (Belopolsky & Theeuwes, 2011).

Some of the earlier studies, investigating working memory (Sperling, 1960; Averbach & Coriel, 1961) used a procedure utilizing stimuli prone to verbal labeling. The authors reported that when requested to answer questions concerning the stimuli, their participants responded with reporting the names of the stimuli verbally. This observation suggested that it is likely that the information of these stimuli was transferred into verbal working memory, rather than to its visual aspect. A following line of research (Palmer, 1988; 1990), attempted to isolate the role of visual attention in the visual aspect of working memory and avoid verbal labeling, by using abstract stimuli such as lines with varying lengths, that are not as easily to name as digits or shapes. The results of these earlier studies provided evidence that visual attention mediates the transfer of information in the visual component of working memory, when verbal working memory is not employed.

More recent works in the field of visual attention and working memory (Schmidt et al., 2002, Theeuwes et al., 2011), approached the problem, using a different strategy to utilize the visual aspect of working memory. In their procedures, participants were requested to perform a concurrent articulatory suppression task, to avoid verbal labeling of the stimuli. More specifically, their participants were requested to repeat aloud a two-digit number in the beginning of each trial, throughout the duration of the entire trial, to ensure that verbal working memory was occupied and visual attention was utilized in the transfer of information to working memory.

At that point, it is worth noting that due to the conditions, in which the current experiments were performed (discussed later on), involving an additional articulatory suppression task was not recommended. Instead, similar to the earlier studies (Palmer, 1988; 1990), the stimuli used in the current procedures consisted of abstract shapes, designed specifically for the purposes of the current work in attempt to avoid verbal labeling and to ensure the involvement of the visual component of working memory.

A considerable line of research, done in the field of visual attention and working memory (Awh & Jonides, 2001; Awh, Jonides, & Reuter-Lorenz, 1998; Vogel, Luck & Shapiro 1998; Belopolsky & Theeuwes, 2009, 2011. Irwin & Gordon, 1998; Schmidt et al., 2002) supports the notion of a close relationship of visual attention with (at least) the visuospatial aspect of working memory, which will be the focus of the current work. Some of the strongest findings are in the field of neuropsychology. Recordings from brain imaging studies, show that there are overlapping frontal and parietal structures involved both in modulating

sensory data in working memory and in performing attention related tasks (Kastner & Pinsk, 2004).

In one of the earlier behavioral studies, the relationship between working memory and visual spatial attention has been empirically tested with a series of experiments, conducted by Awh and colleagues (1998). In their studies, the visual processing advantage was tested with a choice reaction time task. Two independent groups of participants were asked to memorize either a spatial (location of a stimulus) or a non-spatial (identity of a stimulus) aspects of an identical stimulus display, while giving a speeded response to a choice stimulus. The results from their Experiment 1 indicated that participants, retaining in their working memory the spatial aspect of the stimuli, were reliably faster to respond to the choice stimuli, when it appeared in a location that was already kept in their working memory, suggesting facilitation of information processing in the remembered location. The facilitated processing observed in the memorized locations, suggests that participants allocate visual spatial attention to locations held in working memory. In a consecutive experiment, in addition to the main task, the authors introduced an intervening task that required shifts of attention to irrelevant for the main task locations. The results from this manipulation showed that spatial working memory performance was impaired when participants were hindered from attending to a memorized location. The findings of these experiments, support the attention-based rehearsal hypothesis (Awh, Smith & Jonides, 1995; Awh, Vogel & Oh, 2006), which postulates that the shifts in spatial visual attention play a key role in the active maintenance of visual representations in working memory.

A more recent study challenges the classical hypothesis of attention-based rehearsal (Belopolsky & Theeuwes, 2009) by providing evidence that in addition to facilitation of information in memorized locations, also an inhibition of information processing in such locations may occur, depending on working memory demands and the nature of the task. Moreover, the findings of their study suggest that facilitation of information processing in memorized locations does not lead to more accurate memory performance. Importantly, the study challenges the functional role of visual attention in the maintenance of spatial working memory representations.

The role of visual attention in working memory was further elaborated by the findings of Theeuwes and colleagues (Theeuwes, Kramer & Irwin, 2011). Along similar lines of the previous findings, suggesting that visual spatial attention is a mechanism that aids

maintenance of visual representations in working memory (Awh et al., 1995, 1998), Theeuwes et al investigated the role of visual spatial attention when retrieving information from working memory. In their procedure, participants were asked to keep four unique color circles in their working memory. The color circles were presented in the four corners of a screen. The participants' task was to remember these colors and after a brief retention interval, in which the original memory array had vanished, to indicate whether a cue color, now presented in the center of the screen, was part of the initial four colors. In some trials, a probe dot was interrupting the memory task, appearing in one of the four corners of the now empty screen. The probe was with equal probability positioned on a location that is either congruent or incongruent to the location of the cue color. Participants were required to also respond to the probe dot, as soon as they detect it.

The authors were mainly interested in the reaction times in response to the probes. They found that when participants were asked to retrieve a color from their working memory, a probe dot, coinciding with the initial location of the retrieved color, was detected reliably faster compared to when the probe was presented in a location, incongruent to the initial color. These results suggest that when selecting information from visual working memory, people tend to allocate their visual attention to a location that contains task-relevant information.

For the sake of the current work, few points need to be noted. While performing the main task, participants were also required to perform a verbal suppression task, to ensure that the color circles are encoded in visual working memory and not as a verbal code. The exhibited reaction time effect seems to depend on it. In a pilot experiment, neglecting a verbal suppression task, the authors did not find the same reliable results. It appears that spatial attention is allocated only when visual working memory is employed. A second compelling finding is that no effect of the probe on the memory accuracy task was reported. In other words, regardless of participants attending to a location, previously occupied by the required information, they did not report more accurately if the color was presented in the initial array. Interestingly, it seems that the task was easy and a ceiling effect on the accuracy was reached (participants responded with overall 93,3% correct responses). These notes will be discussed in more detail later on.

Taking these findings into account, the proposed conclusion was that *“not only is attention the mechanism by which information is stored in working memory, it is also the mechanism by which information is retrieved by visual working memory”* (Theeuwes, Kramer & Irwin, 2011)

Moving forward, it is intuitively plausible to consider eye movements as a viable component in the interaction between visual attention and working memory. Previous research has shown a tight link between visual attention and eye movements (Deubel & Schneider, 1996; Hoffman & Subramaniam, 1995; Kowler, Anderson, Doshier & Blaser, 1998; Godijn & Theeuwes, 2004). Wherever attention is allocated, typically a saccade is to follow, and the opposite, visual attention is assigned to potential saccade targets (Godijn & Theeuwes, 2004). Moreover, it appears that there is also a direct link between working memory and eye movements. A related study (Belopolsky & Theeuwes, 2011) showed that the eyes deviate away from a remembered stimulus that is not relevant to the main task. Participants were asked to keep the location of a previously presented stimulus in their working memory, while executing an eye movement to a predefined target. Analysis of saccade curvature showed that participants' eye movements deviated away from the remembered location, suggesting that the occupied location, competing with the main task, resulted in the eyes deviating away from that location. These studies provide evidence of a direct link between attention, working memory and the oculomotor system.

This chapter briefly outlined some of the research done in the field of working memory, visual attention and eye movements. As it became apparent, visual attention, working memory and eye movements seem to be tightly linked to each other, creating a well operating chain of processes, involved in our everyday lives. To go one step further, it may be fruitful to view things from a different perspective. More specifically, this chain of processes will be investigated in the context of the phenomenon referred to as looking at nothing, as it can provide invaluable insights of the nature of their relationship.

LOOKING AT NOTHING

Recordings of eye movements in the laboratory show that sometimes people will fixate on blank locations, where relevant information has previously been presented. An example from our daily life, in which this behavior occurs is that sometimes, when we refer to an object that is no longer present in our environment, we may spontaneously look to its previous location despite the fact that we know it's not currently there. This phenomenon is often referred to as looking at nothing and is of considerable interest to various fields, including psycholinguistics, neuropsychology and cognitive science. The concept of looking at nothing still echoes debates among researchers and is suggested to be a valuable asset in the exploration of theories of internal representations, situated cognition (Collins et al., 1989) and embodiment (Barsalou et al., 2003). Furthermore, this phenomenon is a candidate to reveal fundamental aspects of memory, attention and eye movements and is adopted for the purposes of the current research.

One of the earlier reports of people fixating on blank locations comes from the work of Richardson and Spivey (2000). Using eye tracker technology, they recorded saccades fixated on locations, devoid of visual information, when people were requested to recall relevant for their task information, that was previously presented in these locations. In one of their experiments (Richardson & Spivey, 2000), participants viewed a computer screen, divided into four squares. Four different faces, each of them articulating a unique utterance, were placed in each square. After the fourth utterance was articulated, the faces disappeared, leaving a blank screen. Participants then heard the utterance alone, required to determine whether the statement they hear is true or false. What the authors found, was that when participants heard the utterance, simultaneously with validating if it's true or false, they spontaneously fixated the relevant square of the now empty screen. In a follow up experiment Spivey and Geng (2001) revealed the same pattern of results. They asked participants to memorize four simple objects, presented in a 2 by 2 grid on a computer screen. After a short encoding stage, the objects disappeared and the screen was left blank. At that point participants were asked different questions concerning some of the features of the objects (what color was the circle, what direction was the arrow tilted etc.).

Even if the screen was clearly blank, the results revealed that, participant's gaze often fixated the squares, occupied by the objects during the encoding stage, for which information was requested in the retrieval stage. It's important to be noted that in both of these experiments, participants were naïve to their purpose. They were not informed that their eye movements will be recorded, but rather they were deceived that the apparatus and calibration process are for monitoring pupil dilation during the encoding stage.

A similar study done by Altman and Kamide (2004), demonstrates the tight link between speech and eye movements using the visual world paradigm. Participants were asked to review a screen, divided into four squares. In each square an object was placed. Subsequently after, the objects were removed and the screen was left blank. Altman and colleagues recorded the eye movements of participants as they heard spoken sentences. Each time they heard a sentence, that contained a word, related to an object, previously presented in one of the four squares of the screen, participants returned their eyes to the relative square.

Interestingly, the above studies report no facilitation of accuracy in memory performance, in the cases where people fixated the relevant blank locations. In other words, regardless of allocating their visual attention to locations, previously occupied by relevant information, memory retrieval for that information was not improved. These results are in line with the findings of Theeuwes (2011).

Based on these findings (Spivey & Geng, 2001; Richardson & Spivey, 2000; Hoover & Richardson, 2008; Altman & Kamide, 2004), researchers hypothesized that people tend to create only sparse internal representations of the visual environment and that the visual system relies largely on the world as an external memory (O'Regan, 1992). According to that view, looking at nothing is the consequence of the activation of a spatial index associated with the mentioned object (Altman & Kamide, 2004). They argue that the reason memory is not facilitated by returning the eyes on the relevant locations, is because this location no longer contains the appropriate information, requested by the task.

Ferreira et al. (2008) challenges this view, by outlining a theory of integrated internal representations of the looking at nothing phenomenon. They argue that people tend to store visual, spatial and auditory information by forming detailed internal representations of the external world. According to them, looking at nothing is the consequence of re-activation of these representations, leading the eye back to previously viewed regions, in

attempt to enhance memory retrieval. They attribute the lack of accuracy facilitation in previous studies to confounding factors in the experimental procedures. More recent investigations of the eye movements to blank locations (Laeng & Teodorescu, 2002; Johansson & Johansson, 2014; Scholz, Mehlhorm & Krems, 2016; Scholz et al., 2015), seems to support this idea.

In one notable study, Johansson and Johansson (2014) conducted an experiment, manipulating participants' gaze direction in four conditions during a memory retrieval stage. In an episodic memory task, participants were asked to memorize a visual scene. In the retrieval stage of the task, where the screen was already blank, they were asked to recall certain details of the visual scene in four different conditions – free viewing of the blank screen, central fixation on the screen, gazing at an area congruent with the location of the object that is being retrieved and gazing at an area incongruent with the location of the object in question. The results from the free viewing condition, replicated previous findings (Spivey & Geng, 2001; Richardson & Spivey, 2000), showing fixations on the relevant blank locations with no effect on memory performance. When participants were asked to maintain central fixation, memory accuracy performance was hindered for all locations. The crucial finding of this experiment was provided by the results from the third and fourth condition. When asked to fixated on an area, congruent with the objects, for which details were required, participants were significantly more accurate at recalling these details, compared to when their gaze was constrained in an area that is incongruent to the location of the object in question. Their results provide empirical evidence of a functional role of gaze position in memory retrieval. Furthermore, this experiment analyzes the nature of information to-be retrieved, as an independent variable – between object vs within object information. Additional insights provided by the results of this experiment is that the facilitation of memory for spatial relationships between objects is more pronounced compared to memory facilitation for within object features.

An independent line of research, examining the looks at nothing (Scholz, Mehlhorm & Krems, 2016; Scholz, Mehlhorm & Krems, 2016; Scholz et al., 2018), has also suggested facilitation in memory retrieval. The functional role of eye movements to blank spaces was examined in the context of verbal memory retrieval (Scholz, Mehlhorm & Krems, 2016), and rule-based decision making (Scholz et al., 2015).

In an attempt to reveal the fundamental mechanisms, underlying the looking at nothing behavior, Scholz, Klichowicz & Krems (2018) conducted an experiment, focusing their interest on the covert in comparison to overt shifts of visual attention, exhibited in this behavior. Participants were asked to memorize verbal information, corresponding to specific locations on a computer screen. During the retrieval stage of the experiment, all of them solved an unrelated visual tracking task, occurring either in a congruent or incongruent location to the corresponding verbal information. Half of the participants were instructed to solve the tracking task by directing their eyes to the distractor and the other half to maintain central fixation and solve the task while shifting their attention covertly. Results from this experiment demonstrated a tight relationship between memory accuracy performance and the location to which participants directed their attention covertly, by indicating improvement in accuracy of recalling information from the congruent location, even if participants' gaze was not directed there, but the location was attended covertly. These findings provide further evidence for the functional role of looks at empty locations in memory retrieval and go one step further, by suggesting that covert shifts of attention might be sufficient for eliciting improvement in memory retrieval performance.

The stability of the looking at nothing effect in terms of practice was also tested by Scholz, Mehlhorn & Bocklisch (2011). Participants heard four different sentences, each of them associated with a different location on a computer screen and presented 12 times. Subsequently after each presentation, one of the sentences was probed on a blank computer screen. Similar to previous reports (Spivey & Geng, 2001; Richardson & Spivey, 2000), participants tended to direct their gaze to the blank location associated with the probed sentence. Moreover, the results revealed that the more trials participants had completed, the less frequently they fixated their gaze in the relevant blank location, suggesting that the looks at nothing occur more profoundly to empty locations in attempt to aid memory retrieval as long as they are not strongly represented in memory representations.

This chapter discussed some of the research conducted in attempt to understand the basic mechanisms of the looks at blank locations. Reviewing all of the different studies done in this field, it becomes evident that there is more than one way to approach the looking at nothing phenomenon. The results from previous research report strong occurrences of these spontaneous saccades to blank locations, but are still inconclusive with regard to their functionality in memory retrieval.

RATIONALE OF THE CURRENT WORK

It has been well established in laboratory environments as well as outside of them, that people often tend to spontaneously look at empty locations when attempting to recall certain features of an object in question. Although these spontaneous saccades to blank locations have been well documented throughout the literature, little is known about their functionality. Nevertheless, the mechanisms behind this behavior and its role in memory retrieval hold promise to shed light on important aspects of the interplay between visual attention, working memory and the oculomotor system.

The earlier studies of the looking at nothing phenomenon, report no improvement of memory retrieval, when people fixate on a blank location previously occupied by relevant information (Spivey & Geng, 2001; Richardson & Spivey, 2000; Hoover & Richardson, 2008; Altman, 2004). The authors of these studies attribute these results in support of the notion of grounded cognition and embodiment, more specifically, to the theory of the world as its own memory (O'Regan, 1992). More recent investigations, revisiting the looking at nothing phenomenon (Laeng & Teodorescu, 2002; Johansson & Johansson, 2014; Scholz, Mehlhorn & Krems, 2016; Scholz et al., 2015), find evidence of facilitory effects on memory retrieval performance from these looks at blank locations and yield a different perspective – that the looks at nothing serves as a mechanism for re-activation of detailed internal representation of the environment. Regardless of the considerable amount of studies in this area, research of the phenomenon is not depleted and questions of the functional link between looking and nothing and memory retrieval still remain unsettled.

The purpose of the current work was to study the potential functionality of looking at nothing behavior in memory retrieval in the context of a go/no-go task. The first challenge in this series of experiments was to create a task, difficult enough, to produce variance in memory accuracy performance, but still keep a manageable procedure. This manipulation would allow for analysis of accuracy in memory retrieval performance, necessary for investigating prospective facilitation of memory from the looks at blank locations. To that end, Theeuwes et al.'s (2011) procedure was adapted and modified for the purposes of the

current work. The modifications of the original procedure consisted of (1) presenting more complex stimuli, to lower accuracy and explore prospective differences; (2) adding a control condition (probe dot appearing in the center of the display), to obtain a better understanding of the spatial dynamics of the allocation of visual attention; and (3) Stimulus onset asynchronies (SOAs) were included as a variable, to investigate the timeframe in which looks at nothing interact with memory retrieval.

The majority of previous research done in the field has focuses on exploring the looking at nothing phenomenon using the visual world paradigm, where participants are asked to recall certain aspects of previously viewed scenery. The looks at blank locations were recorded using eye tracker technology and performance of memory retrieval was considered. In this current study, reaction time to a probe dot was adapted for the purpose of detecting allocation of visual attention to a specific location of the screen. The performance of memory retrieval will be considered as an important aspect of the current work and memory accuracy will be emphasized in the procedure and analysis.

EXPERIMENT 1

PURPOSE

The purpose of Experiment 1 was to investigate potential allocation of visual attention to blank locations and their effect on memory performance. In Theeuwes et al. (2010) study, allocation of visual attention to the corners of the screen was indicated by reaction times to respond to a probe dot, presented in one of the corners of the screen - either in a congruent or incongruent to the cue stimulus's original location in the memory array. Based on their results it was expected that participants exhibit looks at blank locations previously occupied by relevant stimuli. A main effect of probe dot location was expected, exhibited by faster reaction time to the probe dot on locations that are congruent to the cue stimuli, compared to incongruent or control locations of the probe dot. Similarly, a main effect of stimulus onset asynchrony (SOA), was anticipated to manifest with faster reaction times on trials with longer SOA of the probe dot, compared to trials with shorter probe dot SOA. An interaction between probe dot congruency and SOA would provide information regarding the time course of potential saccades to blank locations. Based on previous studies investigating the looking at nothing phenomenon (Johansson and Johansson, 2013; Scholz et al., 2017), allocation of visual attention to blank locations, previously occupied by relevant information will have a facilitatory effect on memory accuracy performance.

PARTICIPANTS

Seventy-two students (65 females) from New Bulgarian University volunteered to take part in the experiment in exchange for course credits. Their age varied between 18-49, with a mean age of 24. All of them had normal or corrected-to-normal vision and were naïve to the purpose of the study. After providing an informed consent for their participation, participants were notified that those with the three highest accuracies will win a prize (Bluetooth headphones). Eleven participants were excluded for exceeding 30% error rates, leaving 54 participants for the analysis.

APPARATUS AND STIMULI

Given the recent limitations provoked by the COVID-19 pandemic, conducting the experiment in a laboratory environment was not possible. Instead, the experiment was distributed online, and the data were collected through the E-Prime GO (Psychology

Software Tools, Pittsburg, PA) platform. Presumably, participants were performing the task in a casual home environment. No information was obtained about the type of display that was used by every participant

In order to investigate potential attentional shifts, in 80% of the trials, participants were required to detect a small white probe dot, appearing with random probability in one of the four corners of the screen or in the center.

For the memory task, stimuli consisted of five distinct abstract shapes (see Figure 1), designed specifically for the purpose of the current experiment. Presented with white color against a black background, positioned in the four corners of the screen, forming a random configuration each trial. The abstract appearance of the stimuli was designed in attempt to utilize encoding in visual working memory and avoid verbal recoding.



Figure 1. Full set of the stimuli used in the memory task

PROCEDURE

Each trial began with a 2000ms presentation of a memory array consisting of four objects, positioned on the four corners of the screen with a fixation cross in the center. Participants were required to memorize these objects. After a 300ms inter stimulus interval (ISI), the fixation cross was replaced with a cue stimulus, that remained present in the center of the screen for 200ms. Upon seeing the cue stimulus, participants had to determine whether it was presented in the initial memory array. Following an SOA of either 400ms or 800ms, on 80% of the trials, a small probe dot was presented with a random probability on a location either (1) congruent (the same location where the cue stimulus was previously presented), (2) incongruent (the location on the opposite corner of the screen), or (3) control (in the center of the screen). At that point, participants were instructed to make a speeded

response by pressing the spacebar, *only* if they see the probe dot *and* they recalled the cue object as part of the initial memory array. The instructions emphasized that no response should be executed if participants do not see the probe dot, or if they see the probe dot, but do not recall the object as part of the initial configuration. Note that at the point where the probe dot was presented, the initial memory array had already been removed for 1200ms or 1600ms, depending on the SOA. After 1000ms of inter trial interval (ITI), the next trial was initiated. Figure 2 provides a schematic illustration of the trial structure in Experiment 1.

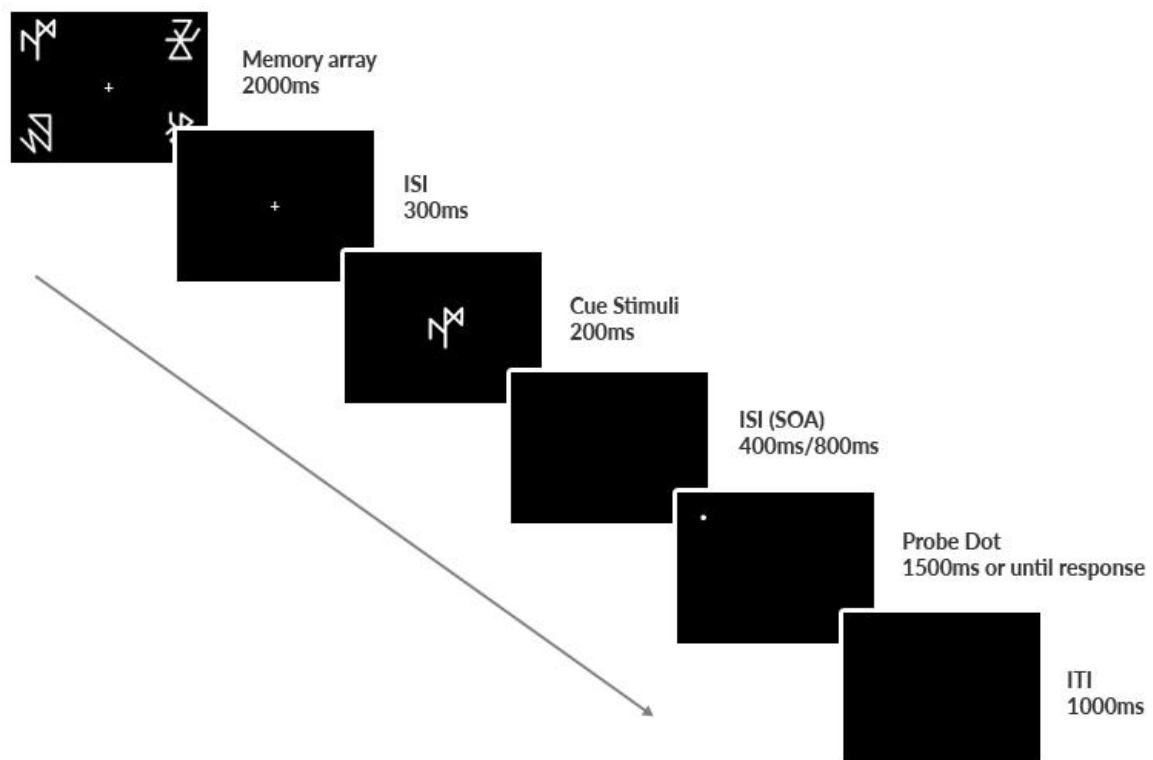


Figure 2. An example of a valid probe dot trial. In this example the probe dot location is congruent to the location of the cue object's previous location.

DESIGN

The experiment constituted a 2x3 within subject design: Stimulus onset asynchrony (400ms vs 800ms) x location of the probe dot (congruent, incongruent or control).

Participants performed a total of 270 experimental trials, preceded by 27 practice trials. In 80% of the experimental trials, a probe dot was presented with equal probability on a

location congruent to the cue object's original location, the opposite corner of the screen or in the center of the screen. In 80% of these probed trials the cue object was part of the initial memory array, making a total of 166 valid trials (where the probe was presented *and* the cue object was part of the initial memory array). In half of all trials the SOA was 400ms and the other half 800ms. Each participant performed 27 trials in each condition.

RESULTS AND DISCUSSION

Accuracy

The data were averaged across subjects. A repeated-measures ANOVA was carried out for the analysis of accuracy, with within-subject factors - condition (location of the probe dot, either congruent vs. control vs. incongruent to the cue object's original location), SOA (400ms vs. 800ms). See table 1 for associated means.

The analysis revealed no significant main effect of condition ($F(2,22) = 1.24, p > .05$). Similarly, no main effect was exhibited by the different SOAs ($F(1,60) = .16, p > .05$). No interaction between the variables was evident from the analysis ($F(2,22) = .30, p > .05$).

Table 1: Experiment 1 mean accuracy (percentage) and (SDs) for each condition

Condition	SOA	
	400	800
Congruent	0,919 (0,07)	0,917 (0,07)
Control	0,924 (0,07)	0,917 (0,07)
Incongruent	0,911 (0,08)	0,913 (0,07)

Reaction Time

Error trials (9,1% of total trials) were excluded from the analysis. The data were analyzed after excluding ± 2 SD from individual subject condition means (resulting in 6,82% loss of the data). The remaining correct RTs were averaged across subjects and entered in a repeated-measures ANOVA. Within-subject factors were considered as follows - condition (location of the probe dot, either congruent vs. control vs. incongruent to the cue object's original position), SOA (400ms vs. 800ms). See table 2 for associated means.

Table 2: Experiment 1 mean RTs (ms) and (SDs) for each condition

Condition	SOA	
	400	800
Congruent	446 (84)	409 (79)
Control	413 (83)	373 (74)
Incongruent	456 (85)	409 (70)

The analysis revealed a main effect of condition ($F(2,22) = 4.65, p < .05, \eta^2 = .44$), with faster RTs when the probe dot was presented in the center of the screen (control condition). Bonferroni post-hoc analysis confirmed that participants were significantly faster to respond to the probe, when it was positioned in the center, but there was no statistically significant difference between the congruent and incongruent positions of the probe. A main effect of SOA was also evident in the analysis ($F(1,60) = 7.70, p < .05, \eta^2 = .56$), revealing significantly faster responses to the probe dot in the longer SOA (800ms). There was no significant interaction between the two factors ($F(2,22) = 1.17, p > .05$). See figure 3.

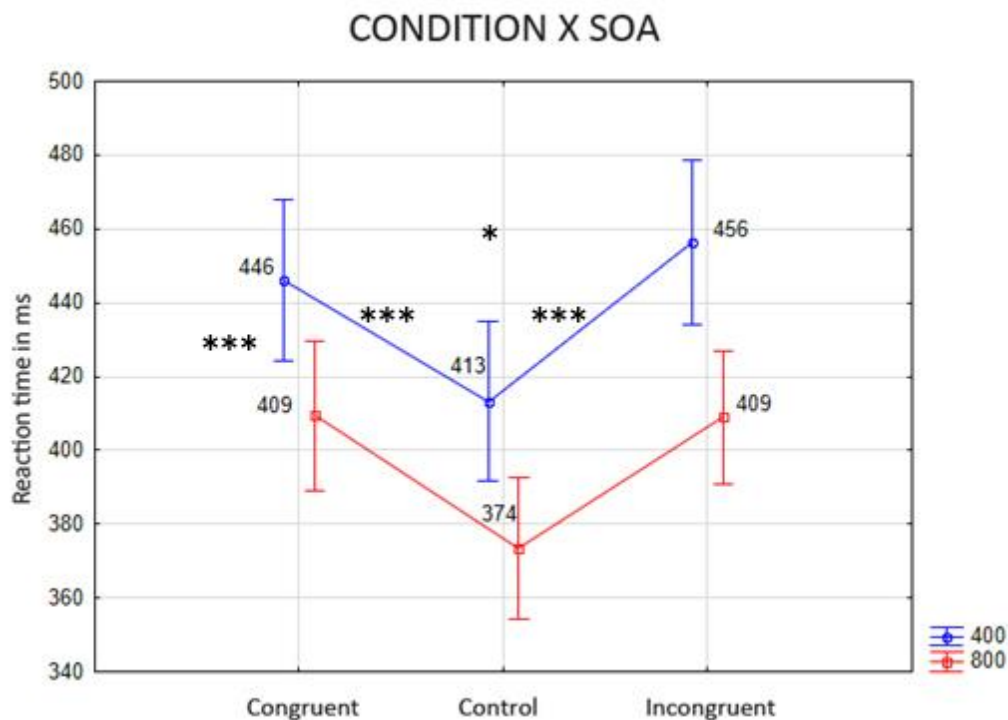


Figure 3: Results from the analysis of condition and SOA variables, subject means. Vertical bars denote standard errors.

Reaction time to detecting the probe dot is essential for the findings of the current study and the main effect of the position of the dot will be addressed first. Although the analysis showed a strong main effect of the position of the probe dot on the screen, it appears that the effect was entirely exhibited in the control condition. Participants were much faster to detect the dot when it was positioned in the center of the screen, compared to when it was positioned either congruent or incongruent to the cue stimulus location. Post-hoc analysis revealed no statistically significant difference between the congruent and incongruent conditions, suggesting that the procedure used in Experiment 1, along with the SOAs of 400ms and 800ms did not obtain any evidence that participants' visual attention may have been cued to the corners of the screen when trying to recall the memory array.

One possible explanation for this effect might be the location of the cue stimulus, presented in the center of the screen. Participants' gaze being fixated in the center of the screen when presented with the cue stimulus, might have resulted in them being faster to detect the probe dot, since it was subsequently presented in the same location. Furthermore, the probe dot appeared in the center of the screen with higher probability compared to any single corner. This was the result of having the same number of control trials, compared to congruent and incongruent trials. As participants realized that the probe dot appears in the center more often than any other position, they may have endogenously maintained attention in the center, thereby hindering allocation of visual attention away from the center of the screen. This possibility will be further elaborated in the next experiment.

The main effect of SOA exhibited faster reaction times to probe dot when it was presented on the screen after the long SOA of 800ms, compared to the short SOA of 400ms. This pattern of results indicates that participants were faster to respond to the probe when they had more time to prepare for it and were expecting for the dot to appear on the screen.

An interaction between the two variables was not found, offering no evidence of saccades occurring to blank locations, or the time course of such oculomotor activity. It is however worth noting that in the short SOA, when the probe dot is presented in a location, congruent with the cue stimulus, participants appear to be slightly faster to detect it (446ms), compared to when it's presented on an incongruent location (456ms). A Neumann-Klaus post-hoc analysis revealed a significant difference between these two points. Although this effect may not be as conclusive evidence of looking at nothing, it may be an indication that looks at blank locations may occur around an SOA of 400ms.

The following studies will explore the possibility that the control condition featuring a probe dot, presented in the center of the screen is interfering with the allocation of visual attention to the corners of the screen. In order to investigate the possibility that looks at nothing occur faster than initially expected, shorter SOAs of the probe dot will be employed in the next experiments.

Analysis of the accuracy from the memory task revealed no effect of the manipulated conditions on participants' memory performance. Considering the absence of allocation of visual attention in the probe detection task, no strong conclusions can be constructed from this analysis.

Nevertheless, this first experiment provided valuable insights for refining the design of the procedure, by addressing the potential issues related to the position and SOA of the probe dot presentation.

EXPERIMENT 2

PURPOSE

The aim of Experiment 2 was to address one of the potential issues that might have contributed to the lack of findings in Experiment 1. As previously discussed, it is plausible that the central position of the probe dot in the control condition, might be involved in the lack of allocation of visual attention away from the center of the screen. To investigate this possibility, the control condition featuring a central probe dot was removed from the design of the current experiment. In the current procedure, each probe dot was located with equal probability in one of the four corners of the screen. As a consequence of this modification, faster reaction times on trials where the probe dot coincides with the cue stimulus' original location on the memory array was expected. Regarding the SOA, the same pattern of results as Experiment 1 was likely to be exhibited, with faster reaction times in the long SOA, compared to the short SOA. An interaction between the two factors would shed light on the time course of looking at nothing effects. Furthermore, facilitation of working memory performance, due to allocation of visual attention to blank spaces, previously occupied by relevant information was anticipated. Such facilitation might exhibit with higher accuracy in memory performance in conditions where the probe dot coincides with the cue object's original location in the memory array.

PARTICIPANTS

Fifty-one students from New Bulgarian University (36 females), aged 19 – 37 (mean age 22) participated in Experiment 2 in exchange for course credits. All of them provided informed consent for their participation and were naïve to the purpose of the study. Nine of the subjects were excluded from the analysis due to low accuracy performance (with a score lower than 70% accuracy), leaving a total of 42 participants.

APPARATUS AND STIMULI

Similar to the first experiment, the current procedure was also distributed online, and data were collected through the E-Prime GO (Psychology Software Tools, Pittsburg, PA) platform. As in Experiment 1, no information was obtained about the type of display that was used by each participant. The stimuli were identical to Experiment 1 (see Figure 1).

PROCEDURE

With regard to the procedure, it was identical to Experiment 1 (see Figure 2), except for the removal of the control condition. In each probe trial, the probe dot was presented in one of the four corners of the screen, in a position either congruent to the cue stimulus's original location in the memory array, or in the opposite corner of the screen. There was no central probe dot employed for this procedure.

DESIGN

The experiment constituted a 2x2 within subject design: Stimulus onset asynchrony (400ms vs 800ms) x location of the probe dot (congruent vs incongruent).

Participants performed a total of 180 experimental trials, preceded by 18 practice trials. In 80% of the experimental trials, a probe dot was presented with equal probability on a location congruent to the cue object's original location or the opposite corner of the screen. In 80% of these probed trials the cue object was part of the initial memory array, making a total of 116 valid trials (where the probe was presented *and* the cue object was part of the initial memory array). In half of all trials the SOA was 400ms and the other half 800ms. Each participant performed 29 trials in each condition.

RESULTS AND DISCUSSION

Accuracy

Repeated measures ANOVA was utilized for the analysis of accuracy after the data were averaged across subjects. Within-subject factors were considered as follows - condition (location of the probe dot, either congruent vs. incongruent to the cue object's original location), SOA (400ms vs. 800ms). See table 3 for associated means.

The analysis revealed no significant main effect of condition ($F(1,42) = .59, p > .05$). Furthermore, no main effect was evident in the two SOAs ($F(1,42) = .05, p > .05$). There was no observed interaction between the two variables ($F(1,42) = .70, p > .05$).

Table 3: Experiment 2 mean accuracy (percentage) and (SDs) for each condition

Condition	SOA	
	400	800
Congruent	0,915 (0,08)	0,919 (0,08)
Incongruent	0,912 (0,07)	0,911 (0,07)

Reaction Time

Error trials (10,2% of total trials) were excluded from the analysis. The data were trimmed under a $\pm 2SD$ cutoff from the individual subject condition means (7,3% of total data removed). The remaining correct RTs were analyzed with a repeated-measures ANOVA, with within-subject factors - condition (location of the probe dot, either congruent vs. incongruent to the cue object's original location), SOA (400ms vs. 800ms). See table 4 for associated means

Table 4: Experiment 2 mean RTs (ms) and (SDs) for each condition

Condition	SOA	
	400	800
Congruent	472 (93)	437 (87)
Incongruent	485 (98)	436 (78)

The data suggested no statistically significant main effect of condition ($F(1,42) = 1.77, p > .05$). Regardless of the position of the probe dot, participants were equally fast to respond to it. A main effect of SOA was again exhibited in the analysis ($F(1,42) = 20.19, p < .05, \eta^2 = .32$). Similar to the first experiment, reaction times to the probe dot were significantly faster in the long SOA of 800ms, compared to the short SOA of 400ms. Analysis of a potential interaction between the two variables revealed no significant interaction between them ($F(1,42) = 2.58, p > .05$). A post-hoc analysis suggested that in the 400ms SOA condition, people were significantly faster to respond to the probe dot, when it coincides with the cue stimulus's original location in the memory array. (see Figure 4).

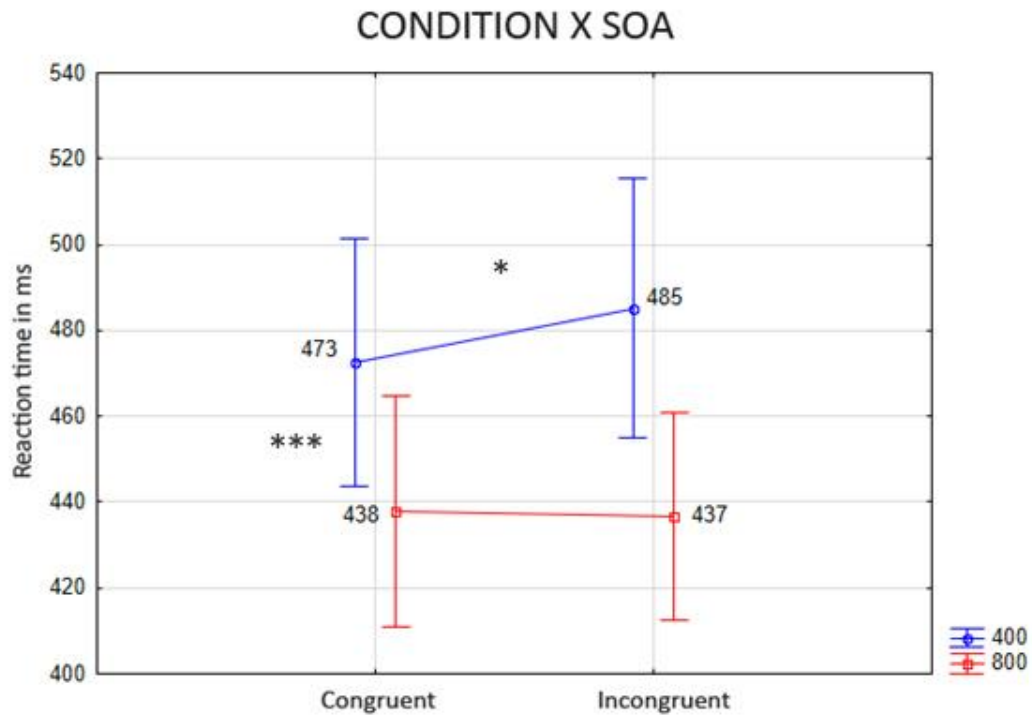


Figure 4: Results from the analysis of condition and SOA variables, subject means. Vertical bars denote standard errors.

The first thing to address is the lack of a main effect on the position of the probe dot. Regardless of the absence of a central probe dot, presumably interfering with the allocation of visual attention away from the center of the screen, it appears that participants' visual attention was not cued to the corners of the screen when trying to recall the memory array. This in turn resulted in no statistically significant difference between reaction times to the probe dot, presented congruently or incongruently to the cue object's original location in the memory array.

Main effect of SOA was again evident in the results of the current experiment. The same pattern as Experiment 1 persisted – reaction times to the probe dot were significantly faster when its onset was 800ms after the cue stimulus, compared to the short SOA of 400ms. Similar to the first experiment's result of the main effect of SOA, the current one can also be explained with the time course of the readiness potential of participants to respond to the probe dot.

The interaction between the position of the probe dot and the SOA was not significant. Similar to Experiment 1, no concrete conclusions about the time course of potential saccades to blank locations can be discussed.

Of particular interest was the participants' reaction times to the congruent positions of the probe dot (472ms) were faster, compared to their reaction times to responding to the probe in the incongruent locations (485ms). As in Experiment 1, post-hoc analysis revealed statistically significant difference between the congruent and incongruent locations of the probe dot, when its onset was 400ms after the cue stimulus disappeared from the screen. This observation is not sufficient to conclude in favor of a looking at nothing effect, however, it provides ground for further considerations of the timing at which potential allocation of visual attention might occur in response to locations previously occupied by relevant stimuli.

At this point, discussion about the results from the analysis of memory performance accuracy might be unproductive. The absence of a firm evidence of saccades to blank locations obscures any possible involvement of that concepts from the discussion. Nevertheless, it might be worth mentioning that regardless of the presence of a marginal effect in the short SOA of 400ms, revealing faster reaction times to the probe dot in locations congruent to the cue object's original location, no such effect is observed in the analysis of accuracy in the same condition.

The next issue that was considered was the time interval in which these saccades to blank locations might be occurring. The results from Experiments 1 and 2, suggested that the looking at nothing effect may be occurring at earlier SOAs. This possibility was investigated in the following experiments.

EXPERIMENT 3

PURPOSE

The reasoning behind the third experiment was related to the time course of potential saccades to blank locations. The analysis of the interaction in Experiments 1 and 2 suggested that looks at nothing may occur at an SOA of 400ms. In contrast, in the long SOA of 800ms, it is plausible that the eyes have already returned to the center of the screen, resulting in no difference between congruent and incongruent probe dot conditions. To investigate this possibility, the current procedure employed shorter SOAs of 100ms (short) and 300ms (long). To be able to directly compare this manipulation to the first experiment, the central probe dot position as a control condition was restored in the current procedure, resulting in a design similar to Experiment 1, with three levels of the probe dot location (congruent vs control vs incongruent). Based on the results of Experiment 1, a main effect of the position of the probe dot was expected, with faster reaction times in the control condition. As opposed to Experiment 1's results, it was anticipated that the effect of the position of the probe dot will extend to the congruent and incongruent conditions, with faster reaction times to the congruent compared to the incongruent probe. Similar to the previous experiments, a main effect of SOA, with faster reaction times to the probe dot in the long SOA (now 300ms), compared to the short SOA (now 100ms) is likely to occur.

PARTICIPANTS

Fifty-four participants (36 females), aged 19 – 44 (mean age 24) were recruited for Experiment 3. Most of them were students from New Bulgarian University and participated in exchange for course credits. All of them provided informed consent for their participation and were naïve to the purpose of the study. Eleven subjects were excluded from the analysis for exceeding 30% error rate.

APPARATUS AND STIMULI

Experiment 3 was also distributed online and data were again collected using the E-Prime GO (Psychology Software Tools, Pittsburg, PA) platform. Participants were presumably performing the task in a casual home environment. As in the previous experiments, no

information was obtained about the type of display that was used by each participant. The stimuli remained unchanged and identical to Experiment 1 (see Figure 1).

PROCEDURE

The procedure was a replication of Experiment 1 (see Figure 2). The sole modification was reduction in the timings of the two SOAs. Instead of SOAs of 400ms and 800ms, the current design employed shorter SOAs of 100ms and 300ms. Everything else remained identical to the procedure of Experiment 1.

DESIGN

The current experiment comprised a 2x3 within subject design: SOA (400ms vs 800ms) x location of the probe dot (congruent vs control vs incongruent).

The design consisted of 270 experimental trials in total, preceded by 27 training trials. In 80% of the experimental trials, a probe dot was presented with equal probability on a location congruent to the cue object's original location, the opposite corner of the screen or in the center of the screen. In 80% of these probed trials the cue object was part of the initial memory array, making a total of 166 trials in which the probe was presented *and* the cue object was part of the initial memory array. In half of all trials the SOA was 100ms and the other half 300ms. Each participant performed 27 trials in each condition.

RESULTS AND DISCUSSION

Accuracy

The accuracy of performance the data were averaged across subjects. Repeated measures ANOVA was again utilized for the analysis, with within-subject factors - condition (location of the probe dot, either congruent vs. control vs. incongruent to the cue object's original location) and SOA (100ms vs. 300ms). See table 5 for associated means.

The analysis obtained no main effect of condition ($F(2,84) = 2.09, p > .05$). No main effect was observed for SOA ($F(1,42) = .34, p > .05$) and no interaction between the two variables ($F(2,84) = .43, p > .05$).

Table 5: Experiment 3 mean accuracy (percentage) and (SDs) for each condition

Condition	SOA	
	100	300
Congruent	0,895 (0,09)	0,894 (0,09)
Control	0,892 (0,09)	0,880 (0,09)
Incongruent	0,874 (0,10)	0,876 (0,09)

Reaction Time

Error trials (12,1% of total trials) were excluded from the analysis. The data were trimmed under a $\pm 2SD$ cutoff from the individual condition means (resulting in total of 6,3% loss of the data). The remaining correct RTs were analyzed with a repeated-measures ANOVA. The current experiment comprised of condition (location of the probe dot, either congruent vs. control vs. incongruent to the cue object's original location) and SOA (100ms vs. 300ms) as within-subject factors. See table 6 for associated means.

Table 6: Experiment 3 mean RTs (ms) and (SDs) for each condition

Condition	SOA	
	100	300
Congruent	549 (123)	489 (115)
Control	521 (126)	451 (110)
Incongruent	551 (130)	499 (122)

The current analysis showed a main effect of condition ($F(2,88) = 17.43, p < .05, \eta p^2 = .28$). Similar to Experiment 1's results, participants were significantly faster to respond to the probe dot, when it was presented in the center of the screen. A post-hoc analysis, suggested no statistically significant difference between the congruent and incongruent positions of the probe, in relation to the cue stimulus's original location in the memory array. Another strong main effect of SOA was also exhibited in this experiment ($F(1,44) = 55.00, p < .05, \eta p^2 = .55$), with reaction times to the probe dot, much faster in the long (300ms) SOA. Similar to the results of the previous experiments, no interaction between the two variables was observed in this experiment ($F(2,88) = 1.04, p > .05$). (See Figure 5).

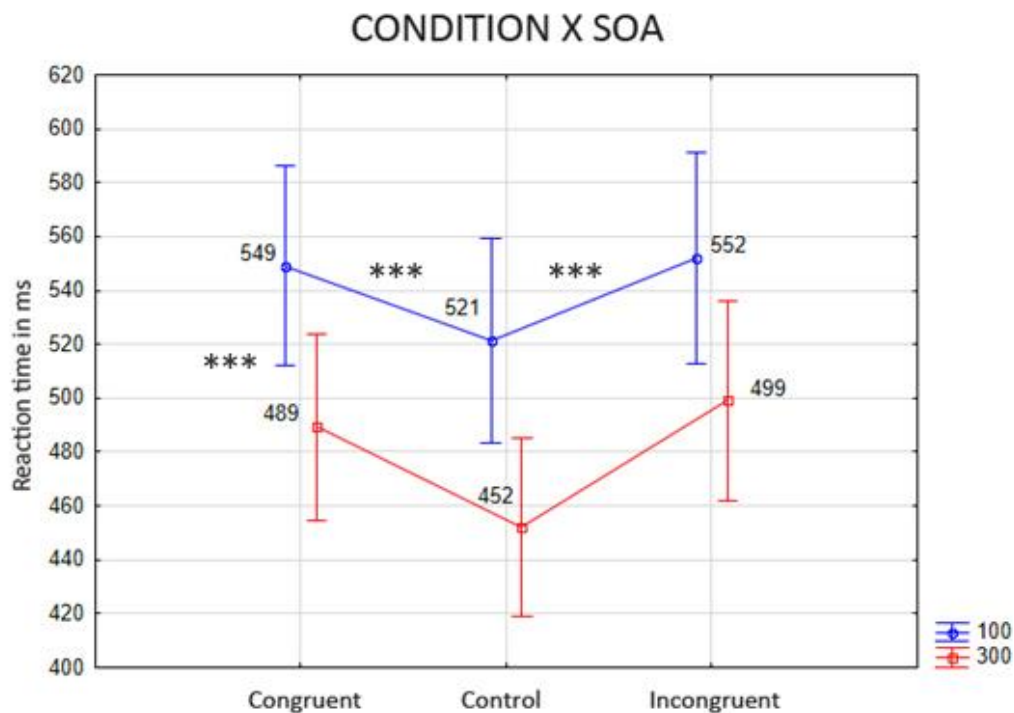


Figure 5: Results from the analysis of condition and SOA variables, subject means. Vertical bars denote standard errors.

A main effect of the position of the probe dot was evident in the analysis. This effect is in line with Experiment 1, again characterized by faster reaction times to the probe in the control condition. What is unexpected in the results from the analysis of the main effect of condition is the lack of statistically significant difference between the congruent and incongruent conditions of the probe. Consistent with the preceding experiments, the current procedure was also not able to produce evidence of cuing participants' visual attention to the corners of the screen.

As expected, participants were significantly faster to respond to the probe dot in the long SOA of 300ms. It appears that the short SOA of 100ms doesn't provide enough time for participants to prepare and be ready to respond to the probe dot, resulting in them being significantly faster to detect the probe and respond to it, when its SOA is longer - in the current case 300ms.

In exploring an interaction between the position of the probe dot and the SOA, the data was consistent with Experiment 1 and 2. No statistically significant interaction between the two variables was evident. It is however worth noting that a post-hoc analysis revealed that

in line with the preceding two experiments, participants were faster to respond to the probe dot, when it was positioned in congruent locations (489ms), compared to incongruent (499ms), when it's SOA was 300ms. Although the effect is not statistically significant, it seems that this trend is observed throughout the series of experiments, provoking further considerations of the time course of potential looks at nothing.

Finally, the analysis of the accuracy data showed no effect in the different conditions. This was to be expected since reaction times did not provide any evidence that attention was allocated to the corners of the screen.

Further investigation will be attempted, utilizing the findings of the preceding experiments. The marginal difference in the reaction times occurring around 300ms appears to be consistent throughout the experiments, suggesting that saccades at blank locations may be occurring around that time. It is plausible consider this possibility in relation to the results of Experiment 2.

EXPERIMENT 4

PURPOSE

The purpose of Experiment 4 was to implement the short SOAs in the design of Experiment 2. The procedure of the current experiment comprised of short SOAs of 100ms and 300ms. Similar to Experiment 2, the control condition featuring a central presentation of the probe dot was removed. Based on these manipulations, allocation of visual attention to the corners of the screen is expected, exhibited by faster reaction times to the congruent, compared to the incongruent probe dot. The conventional effect of SOA is also expected to occur, exhibited by faster reaction times to the probe dot in the long SOA of 300ms. Interaction between the two variables is plausible, revealing the timeframe in which potential shifts of attention to blank locations might occur. In the event of successful cuing of participants attention to the corners of the screen, facilitation of memory performance, manifesting with higher accuracy in the congruent, compared to the incongruent condition is anticipated.

PARTICIPANTS

Fifty-one participants (45 females), aged 19 – 46 (mean age 24) were recruited for Experiment 4. Most of them were students from New Bulgarian University and participated in exchange for course credits. All of them were naïve to the purpose of the current study and provided informed consent before attempting to participate in the experiment. Ten subjects were excluded from the analysis for exceeding 30% error rate.

APPARATUS AND STIMULI

Similar to the preceding experiments, Experiment 4 was also distributed online and data were again collected using the E-Prime GO (Psychology Software Tools, Pittsburg, PA) platform. As in the previous experiments, no information was obtained about the type of display that was used by each participant. The stimuli remained unchanged and identical to Experiment 1 (see Figure 1).

PROCEDURE

The procedure was a replication of Experiment 2 (see Figure 2). The following modifications were implemented: (1) removing the control condition. In each probe trial, the probe dot

was presented in one of the four corners of the screen, in a position either congruent to the cue stimulus's original location in the memory array, or in the opposite corner of the screen. Similar to Experiment 2, no central probe dot was employed in the current procedure. (2) Instead of SOAs of 400ms and 800ms, the current design employed shorter SOAs of 100ms and 300ms.

DESIGN

The current experiment comprised a 2x2 within subject design: SOA (100ms vs 300ms) x location of the probe dot (congruent vs incongruent).

Participants performed a total of 180 experimental trials, preceded by 18 practice trials. In 80% of the experimental trials, a probe dot was presented with equal probability on a location congruent to the cue object's original location or the opposite corner of the screen. In 80% of these probed trials the cue object was part of the initial memory array, making a total of 116 valid trials (where the probe was presented *and* the cue object was part of the initial memory array). In half of all trials the SOA was 400ms and the other half 800ms. Each participant performed 29 trials per condition.

RESULTS AND DISCUSSION

Accuracy

Repeated measures ANOVA was utilized for the analysis of accuracy after the data were averaged across subjects. Within-subject factors were considered as follows - condition (location of the probe dot, either congruent vs. incongruent to the cue object's original location), SOA (100ms vs. 300ms). See table 7 for associated means.

The analysis revealed no significant main effect of condition ($F(1,42) = .20, p > .05$). No main effect of SOA was evident ($F(1,42) = .18, p > .05$) and no interaction was observed between the two variables ($F(1,42) = .22, p > .05$).

Table 7: Experiment 4 mean accuracy (percentage) and (SDs) for each condition

Condition	SOA	
	100	300
Congruent	0,907 (0,08)	0,900 (0,10)
Incongruent	0,900 (0,08)	0,900 (0,09)

Reaction Time

Error trials (9,8% of total trials) were excluded from the analysis. The data were trimmed under a $\pm 2SD$ cutoff from the condition means (resulting in total of 6,2% loss of the data). The remaining correct RTs were analyzed with a repeated-measures ANOVA. The remaining correct RTs were analyzed with a repeated-measures ANOVA, with within-subject factors - condition (location of the probe dot, either congruent or incongruent to the cue object's original location) and SOA (100ms vs. 300ms).

Table 8: Experiment 4 mean RTs (ms) and (SDs) for each condition

Condition	SOA	
	100	300
Congruent	524(115)	454(102)
Incongruent	516(106)	455(89)

The analysis of the data from Experiment 4 showed no main effect of condition ($F(2,88) = .35, p > .05$). Regardless of the location of the probe dot, the results suggest that there were no differences in the reaction times to respond to it. In line with the previous experiments, the conventional strong main effect on SOA was also evident in this experiment ($F(1,44) = 68.00, p < .05, \eta p^2 = .55$), exhibiting with reaction times to the probe dot, much faster in the currently long SOA of 300ms. An interaction between the two variables was not observed in the current analysis ($F(2,88) = .78, p > .05$). (See Figure 6).

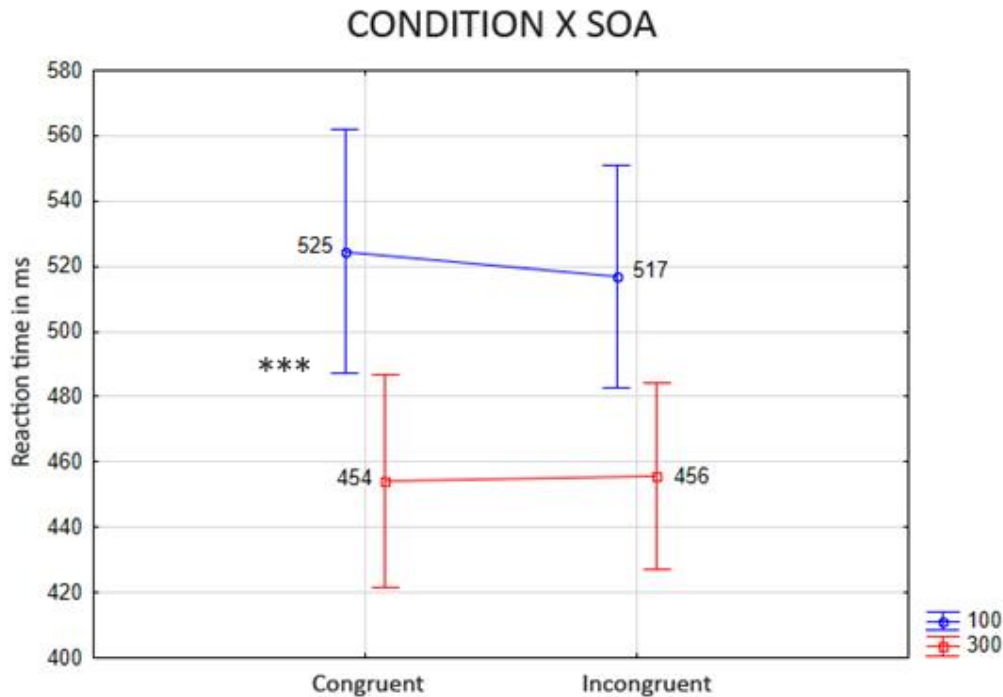


Figure 6: Results from the analysis of condition and SOA variables, subject means. Vertical bars denote standard errors.

Reaction times to the probe dot in the different locations of the screen is fundamental for the current study and the results from the analysis of the condition variable will be discussed first. In Experiment 4, there was no observed main effect of the location of the probe dot in relation to the cue stimulus's original location in the memory array. Regardless of the absence of a control condition, it appears that when trying to recall the memory array, participants' visual attention was not cued to the corners of the screen and their reaction time was not influence by the congruent or incongruent location of the probe dot. This result is in line with the preceding three experiments.

Similar to the previous experiments, the strong main effect of SOA was also evident in the current results, again exhibiting with reaction times to the probe dot significantly faster in the 300ms SOA condition, compared to the short SOA of 100ms. The same explanation for this effect, as in Experiment 3 is plausible. When the probe is presented only 100ms after the cue stimulus has withdrawn from the screen, participants haven't had enough time to prepare and be ready to respond to the probe dot, resulting in them being significantly faster to detect the probe and respond to it in the longer onset of 300ms.

Consistent with the preceding experiments, an interaction between the location of the probe dot in relation to the cue stimulus's original location and the SOA was not found, indicating that regardless of when the probe dot was presented on the screen, there was no significant difference in participants' reaction times to respond to it, in relation to its position on the screen. Furthermore, the previously observed marginal effect in the reaction times of responding to the probe, between its congruent and incongruent conditions in the long SOA (300ms.), observed consistently throughout the first three experiments was not exhibited in Experiment 4.

Finally, the analysis of accuracy of memory performance showed no difference in the accuracy in any of the conditions. Given the lack of apparent shifts of attention to the corners of the screen, no valuable conclusions can be extracted by this result.

ADDITIONAL ANALYSES

The present study consisted of four experiments, designed for the purpose of detecting potential looking at nothing behavior and measuring its potential implications in memory accuracy performance. The results of the experiments did not reveal any consistent evidence of allocation of visual attention to the corners of the display during recall. To address certain issues considered as plausible factors behind the lack of expected results, additional analyses of the data from the four experiments were performed. More specifically, (1) the individual performance of the participants, (2) possible practice effects and (3) potential effects induced by the spatial dynamics of the position of the probe dot, were considered as factors that may provide additional insights into the mechanisms involved in the experiments. To that end, an analysis of the best performing participants, a correlational analysis between effect size and accuracy, block analysis and probe dot location analysis were carried out.

BEST PERFORMING PARTICIPANTS ANALYSIS

Conducting the experiments in a laboratory environment was not an option during the COVID-19 pandemic, instead they were distributed online and presumably participants were performing the task in an uncontrolled, casual environment. In order to explore the possibility that these conditions affected participants' focus while performing the task, which might have contributed to the lack of looking at nothing effects in the results of the experiments, an analysis of the best performing participants was carried out. Reaction time to the probe dot data from the top 30% participants that obtained the highest accuracy in the memory task were analyzed, in attempt to investigate whether allocation of visual attention to the corners of the display manifests, when participants were focused and performed the task diligently.

Experiment 1

Data from 18 participants were analyzed. All of them had memory accuracy performance higher than 85%. A repeated measures ANOVA was carried out for the analysis with within-subject factors – location of the probe dot (congruent vs incongruent vs control), SOA (400ms vs 800ms).

The analysis revealed a statistically significant difference in the main effect of condition ($F(2,22) = 21.9, p < .05, \eta^2 = .56$), with reaction times to the probe dot faster in the control condition and no significant difference between the congruent and incongruent conditions. A main effect of SOA was also suggested by the analysis ($F(1,60) = 67.7, p < .05, \eta^2 = .79$), with reaction times to the probe faster in the 800ms compared to the 400ms SOA.

There was no statistically significant interaction between the two factors ($F(2,22) = 2.13, p > .05$). The results of the analysis of the best performing participants from Experiment 1 provided no evidence of allocation of visual attention to the corners of the screen.

Experiment 2

A repeated measures ANOVA was carried out for the analysis of the best performing 16 participants in Experiment 2, with within-subject factors location of the probe dot (congruent vs incongruent) and SOA (400ms vs 800ms).

No significant difference in the main effect of condition was observed ($F(2,22) = 1.68, p > .05$). A main effect of SOA was exhibited ($F(1,60) = 8.4, p < .05, \eta^2 = .35$), with reaction times to the probe faster in the 800ms compared to the 400ms SOA. Interestingly, the analysis of the best performing participants from Experiment 2 showed a statistically significant interaction between the location of the probe dot and the SOA ($F(2,22) = 9.05, p < .05, \eta^2 = .37$) with reaction times to the probe dot faster when it's location coincided with the cue stimulus' original location and its SOA was 400ms. (See Figure 7).

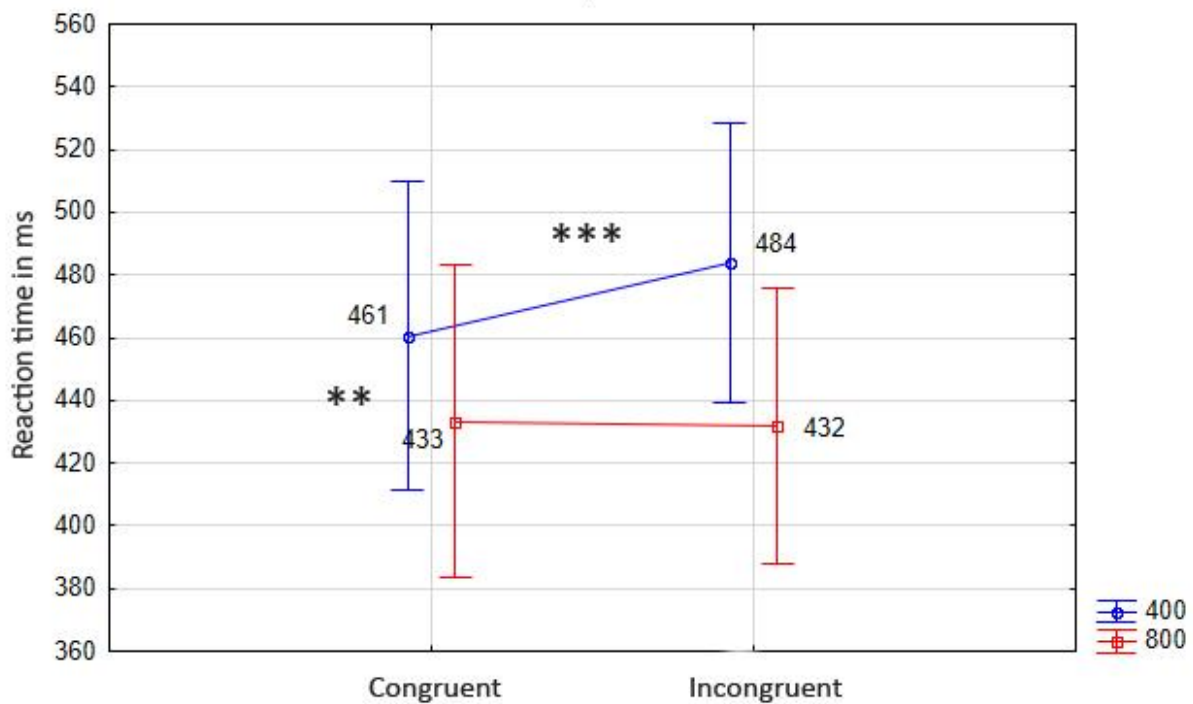


Figure 7: Results from the analysis of condition and SOA variables, subject means. Vertical bars denote standard errors.

Experiment 3

Data from Experiment 3's best performing 14 participants were analyzed with a repeated measures ANOVA with within-subject factors as follows: location of the probe dot (congruent vs control vs incongruent) and SOA (100ms vs 300ms).

The main effect of condition was significant in the analysis ($F(2,22) = 11.2, p < .05, \eta^2 = .46$), but showed no significant difference between the congruent and incongruent conditions (Only the control condition was different). A main effect of SOA was also exhibited ($F(1,60) = 34.4, p < .05, \eta^2 = .72$), with reaction times to the probe faster in the 300ms compared to the 100ms SOA. There was no statistically significant interaction between the two factors ($F(2,22) = 1.57, p > .05$).

Experiment 4

Data from 13 participants were analyzed with a repeated measures ANOVA with within-subject factors – location of the probe dot (congruent vs incongruent) and SOA (100ms vs 300ms).

No significant difference in the main effect of condition was observed ($F(2,22) = 0.51, p > .05$). A main effect of SOA was exhibited in the analysis ($F(1,60) = 22.7, p < .05, \eta^2 = .65$), with reaction times to the probe faster in the 300ms compared to the 100ms SOA. There was no statistically significant interaction between the two factors ($F(2,22) = 0.03, p > .05$).

The analysis of the best performing participants did not reveal solid evidence of looks at blank locations from participants with high memory accuracy performance in Experiments 1, 3 and 4. It is worth noting that in Experiment 2, a statistically significant interaction between the two factors was observed, providing some evidence that in the 400ms SOA, allocation of visual attention to the corners of the screen was exhibited by participants with high memory accuracy performance. In the context of the lack of such effect in the rest of the experiments, this significant interaction was inconclusive as to whether high accuracy performance and looks at blank locations are correlated. A different approach to exploring this possibility is presented in the next section.

CORRELATION ANALYSIS

To further explore the possibility that individual participants exhibit a potential looking at nothing effect with a different magnitude based on their memory accuracy performance, a correlation analysis was performed. Data from all participants that scored higher than 75% accuracy were analyzed. To calculate the magnitude of the looking at nothing effects, the individual reaction times to the probe dot in the congruent locations of the probe were subtracted from the reaction times in the incongruent locations. A positive magnitude of the effect would offer evidence for allocation of visual attention to the corners of the screen. Across the four experiments, only the 300 and 400ms SOA conditions were included in the correlational analysis as previous results had suggested that looking at nothing effects were more likely to be observed around these SOAs.

Results

The results revealed no statistically significant correlation between the individual participants' memory performance score and the magnitude on the effects in terms of reaction time to the probe dot.

Experiment 1 - ($r = 0.134, p > .05$).

Experiment 2 - ($r = -0.060, p > .05$).

Experiment 3 - ($r = -0.217, p > .05$).

Experiment 4 - ($r = 0.163, p > .05$).

BLOCK ANALYSIS

Scholz et al. (2011) reported a practice effect of the looking at nothing phenomenon. In their study, the authors observed that the more trials participants had completed, the less frequently they fixated their gaze in the relevant blank locations, concluding that the looking at nothing effect diminishes with practice. It is plausible to consider this practice effect as a contributing factor to the lack of looking at nothing behavior observed in the current work. Another possibility that will be explored in the following analysis is that throughout the duration of the experiments, participants became familiar with the stimuli and labeled them verbally. Previous studies (Schmidt et al., 2002; Theeuwes et al., 2011) reported that when the stimuli are coded in the verbal component of working memory, visual working memory is not employed, thus hindering the allocation of visual attention to the corners of the screen. To that end, a block analysis was performed with each of 4 blocks containing 25% of trials, allowing for a more detailed analysis of the reaction times to the probe dot in the time course of each experiment.

Results

A repeated measures ANOVA was carried out for the block analysis with within-subject factors as follows:

Experiment 1 - location of the probe dot (congruent vs incongruent vs control), SOA (400ms vs 800ms) and block (block 1: 0-75 trials, block 2: 75-150 trials, block 3: 150-225 trials, block 4: 225-297 trials).

Experiment 2 - location of the probe dot (congruent vs incongruent), SOA (400ms vs 800ms) and block (block 1: 0-50 trials, block 2: 50-100 trials, block 3: 100-150 trials, block 4: 150-197 trials).

Experiment 3 - location of the probe dot (congruent vs incongruent vs control), SOA (100ms vs 300ms) and block (block 1: 0-75 trials, block 2: 75-150 trials, block 3: 150-225 trials, block 4: 225-297 trials).

Experiment 4 - location of the probe dot (congruent vs incongruent), SOA (100ms vs 300ms) and block (block 1: 0-50 trials, block 2: 50-100 trials, block 3: 100-150 trials, block 4: 150-197 trials).

The analyses revealed no statistically significant three-way interaction with block as a factor in any of the four experiments.

Experiment 1 - ($F(2,22) = 1.07, p > .05$). Experiment 2 - ($F(2,22) = 1.18, p > .05$).

Experiment 3 - ($F(2,22) = 1.59, p > .05$). Experiment 4 - ($F(2,22) = 0.65, p > .05$).

Unlike previous research (Scholz et al., 2011) the block analysis of the current experiments provided no evidence for practice effects in any of them. The data revealed no results that support the possibility that participants became familiar with the stimuli in the process of performing the task and labeled them verbally.

PROBE DOT POSITION ANALYSIS

Another question of interest is the possibility that participants exhibited a natural tendency to look at a particular corner of the screen during stimulus recall. It is plausible to consider that if such a tendency exists, reaction times to the probe dot will be faster, when the probe is presented in that particular location on the screen. This effect may have contributed to biasing the allocation of visual attention to a specific corner of the screen, and, to an extent, overriding looking at nothing effects to other locations on the display.

Results

The position of the probe dot was analyzed with a repeated-measures ANOVA, with within-subject factors - location of the probe dot (Experiments 1 and 3: congruent vs incongruent vs control; Experiments 2 and 4: congruent vs incongruent), SOA (Experiments 1 and 2: - 400ms vs 800ms; Experiments 3 and 4: - 100ms vs 300ms) and position of the probe dot (bottom left vs bottom right vs top left vs top right).

There was no statistically significant three-way interaction observed with position of the probe dot as a factor in any of the four experiments.

Experiment 1 - ($F(2,22) = 0.17, p > .05$). Experiment 2 - ($F(2,22) = 1.02, p > .05$).

Experiment 3 - ($F(2,22) = 0.83, p > .05$). Experiment 4 - ($F(2,22) = 0.25, p > .05$).

GENERAL DISCUSSION

Over the past few decades, studies focused on visual attention have recorded spontaneous saccades to blank locations, when participants were requested to recall certain features of an object, no longer available for inspection. This phenomenon referred to as looking at nothing is consistently reported throughout the literature and is considered to be a valuable asset in the exploration of the functional interaction between visual attention, working memory and the oculomotor system. Some of the earlier studies (Spivey & Geng, 2001; Richardson & Spivey, 2000; Hoover & Richardson, 2008; Altman, 2004) report that regardless of participants fixating their eyes on blank locations, previously occupied by relevant stimuli for which they require information, facilitation of memory accuracy was not evident. The findings of these earlier studies were challenged by Ferreira et al., 2008. They contributed this lack of facilitation in memory accuracy to methodological issues and proposed that the looks at blank locations serve as a mechanism for re-activation of detailed internal representations of the environment and have a facilitatory effect on memory performance. Later on, empirical evidence of a functional role of visual attention in working memory was provided by Johansson and Johansson (2014), suggesting that participants' memory accuracy performance was facilitated by the looks at blank locations, previously occupied by relevant information. These results have later been supported by an independent line of research, conducted by Scholz and colleagues (Scholz, Mehlhorn & Krems, 2016; Scholz, Mehlhorn & Krems, 2016; Scholz et al., 2018). Using different methods, Theeuwes et al. (2010), found that allocation of visual attention to a blank location on the screen, previously occupied by a relevant stimulus, did not enhance memory accuracy performance. The inconsistency in the results from these studies suggests that research of the looking at nothing phenomenon is not depleted and further investigations of the interaction between visual attention and working memory are necessary to assess potential functionality between them.

The current study employed a procedure, similar to Theeuwes et al. (2011) in attempt to investigate the potential functional role of visual attention in working memory. In a series

of experiments, participants were requested to hold a number of objects in their working memory while also performing a simple detection task. After memorizing the objects from the initial memory array, a cue stimulus appeared in the center of the screen. Participants were requested to respond whether the cue stimulus was part of the initial configuration. While retrieving this information, on some trials, a probe dot, prompting a speeded response, was presented on the blank screen, in a location congruent or incongruent to the cue stimulus' original location in the memory array. The rationale behind this manipulation was that if participants are faster to detect the probe dot in a location, congruent to the cue stimulus' original location, this reaction time effect will provide evidence for allocation of visual attention to this location in line with the looking at nothing literature. Stimulus onset asynchrony of the probe dot was also manipulated in the current study, providing each procedure with two different levels of SOA, in attempt to reveal the time at which potential fixations to blank location might influence task performance. It was expected that the results will reveal a main effect of the location of the probe dot, with faster reaction times for locations, congruent to the cue stimulus's original location in the memory array. A main effect of SOA with faster reaction times in the longer SOAs was also anticipated. An interaction between the two variables would have provided evidence for the time at which looking at nothing effects occur in the procedure.

The results obtained across the four experiments did not reveal any evidence of looking at nothing behavior. In other words, participants were not significantly faster to detect the probe dot when it was presented in a location, congruent with the cue stimulus' original location in the memory array. These results suggest that visual attention was not cued to the corners of the screen during stimulus recall. Possible reasons for the inability to obtain evidence looking at nothing are discussed below.

One factor to consider is the conditions in which participants were performing the task. Given the recent limitations provoked by the COVID-19 pandemic, conducting the experiments in a laboratory environment was not an option. Instead, the experiment was distributed online, and the data were collected through the E-Prime GO platform. There was no information obtained about the type and size of monitors on which the procedure was viewed and presumably these parameters varied across the subjects. It is possible that this uncontrolled variation resulted in differences in the visual angle and subject positioning relative to the display and is possibly a factor that influenced the results of the experiments.

Furthermore, participants were performing the task in a casual home environment with no supervision by an experimenter, as opposed to the controlled environment of the laboratory, devoid of any distractions, suitable for the experimental purposes. It is possible that these conditions affected participants' involvement in the task, resulting in an absence of focus while performing it. The average accuracy performance was relatively low, which might also be considered in support for a potential issue with participants' motivation to perform the task. To investigate this possibility, an analysis of the best performing participants was carried out. The results of the analysis revealed that only in Experiment 2, there was some evidence of a looking at nothing behavior exhibited by the best performing participants, with reaction times to the probe significantly faster in its congruent locations in the 400ms SOA. Having in mind that there was no such pattern of results in the rest of the experiments, the observed effect in Experiment 2 is inconclusive. Furthermore, a correlation analysis ruled out the possibility of a positive correlation between task accuracy and looking at nothing effect size in all four of the experiments.

Another point worth considering was noted in the introduction of the current work. The procedure of Theeuwes et al. (2011), which inspired the current one, also involved a verbal suppression task. While performing the main task, participants were also required to read aloud a two-digit number, presented in the center of the screen in the beginning of each trial, throughout the duration of the trial. The reason for this additional task was to ensure that the stimuli (color circles) are encoded in visual working memory and not as a verbal code. The authors reported that in a pilot experiment, not involving a verbal suppression task, there was no sign of allocation of visual attention to the corners of the screen. According to them, visual spatial attention is allocated only when visual working memory is employed. Due to the difficulty of conducting an additional verbal suppression task, given the previously mentioned conditions in which the main task was performed (casual home environment where supervision of participants while performing the task was not possible), the procedures of the current experiments did not involve any form of verbal suppression. Instead, the stimuli were specifically designed for the purposes of the current study. They consisted of five distinct abstract shapes. Although, the stimuli were designed so that they are not prone to verbal labeling, it is possible that in the process of performing the task participants became familiar with the stimuli and labeled them using some form of a verbal code. In this event, visual working memory would most likely not be employed to the

degree necessary to induce looking at nothing effects. In order to explore the potential familiarization with the stimuli throughout the duration of the experiment that may have hindered participants' visual attention to the corners of the screen, a block analysis was carried out. Each experiment was divided into four blocks in an attempt to provide a more detailed picture of the reaction times to the probe dot in the time course of the experiment. The analysis showed that participants were equally fast to detect the probe, regardless of any practice effects, suggesting that familiarization with the stimuli and procedure were not a significant factor that influenced potential looking at nothing behavior.

The possibility that participants exhibited a natural tendency to allocate visual attention to a particular corner of the screen while retrieving information about the stimuli was also investigated in the preceding section. An analysis of the location of the probe dot was performed in an attempt to explore a possible overriding of potential looking at nothing effects by a bias of participants' visual attention to a specific location of the display. The analysis of the probe dot location revealed no evidence in support for that possibility. The reaction times to the probe were not significantly different between the four corners of the display, suggesting that participants' visual attention was not biased to a particular corner of the screen during stimulus recall.

Finally, there are some observations in the experiments that are worth mentioning. Although not considered as a solid evidence of a looking at nothing behavior, the results of Experiments 1, 2 and 3 revealed an indication of allocation of visual attention to the corners of the screen. It appears that participants were slightly faster to respond to the probe dot, when it was presented on locations congruent to the cue stimulus' original location in the 300ms and 400ms SOAs. This difference was not large enough to produce an interaction between the SOA and the location of the probe dot, but a post-hoc analysis confirmed that the difference between these two points is significant. This observation was not consistent and hence, inconclusive for the purposes of the current work, but it may suggest that looking at nothing effects occur around 300ms and decay by 800ms. Future studies employing the current procedure should focus on the 300ms and 400ms SOA range and include a verbal suppression task in order to be more effective in eliciting looking at nothing effects.

REFERENCES

- Altmann, G. T. (2004). Language-mediated eye movements in the absence of a visual world: The 'blank screen paradigm'. *Cognition*, 93(2), B79-B87.
- Awh, E., Jonides, J., & Reuter-Lorenz, P. A. (1998). Rehearsal in spatial working memory. *Journal of Experimental Psychology: Human Perception and Performance*, 24(3), 780.
- Awh, E., & Jonides, J. (2001). Overlapping mechanisms of attention and spatial working memory. *Trends in cognitive sciences*, 5(3), 119-126.
- Awh, E., Vogel, E. K., & Oh, S. H. (2006). Interactions between attention and working memory. *Neuroscience*, 139(1), 201-208.
- Baddeley, A. D., & Hitch, G. (1974). Working memory. In *Psychology of learning and motivation* (Vol. 8, pp. 47-89). Academic press.
- Baddeley, A. D., & Logie, R. H. (1999). Working memory: The multiple-component model.
- Barsalou, L. W., Niedenthal, P. M., Barbey, A. K., & Ruppert, J. A. (2003). Social embodiment.
- Belopolsky, A. V., & Theeuwes, J. (2009). Inhibition of saccadic eye movements to locations in spatial working memory. *Attention, Perception, & Psychophysics*, 71(3), 620-631.
- Belopolsky, A. V., & Theeuwes, J. (2009). No functional role of attention-based rehearsal in maintenance of spatial working memory representations. *Acta Psychologica*, 132(2), 124-135.
- Belopolsky, A. V., & Theeuwes, J. (2011). Selection within visual memory representations activates the oculomotor system. *Neuropsychologia*, 49(6), 1605-1610.
- Brown, J. S., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. *Educational researcher*, 18(1), 32-42.
- Deubel, H., & Schneider, W. X. (1996). Saccade target selection and object recognition: Evidence for a common attentional mechanism. *Vision research*, 36(12), 1827-1837.

- E. Irwin Robert D. Gordon, D. (1998). Eye movements, attention and trans-saccadic memory. *Visual cognition*, 5(1-2), 127-155.
- Ferreira, F., Apel, J., & Henderson, J. M. (2008). Taking a new look at looking at nothing. *Trends in cognitive sciences*, 12(11), 405-410.
- Godijn, R., & Theeuwes, J. (2004). The relationship between inhibition of return and saccade trajectory deviations. *Journal of Experimental Psychology: Human Perception and Performance*, 30(3), 538.
- Hoffman, J. E., & Subramaniam, B. (1995). The role of visual attention in saccadic eye movements. *Perception & psychophysics*, 57(6), 787-795.
- Hoover, M. A., & Richardson, D. C. (2008). When facts go down the rabbit hole: Contrasting features and objecthood as indexes to memory. *Cognition*, 108(2), 533-542.
- Johansson, R., & Johansson, M. (2014). Look here, eye movements play a functional role in memory retrieval. *Psychological science*, 25(1), 236-242.
- Kastner, S., & Pinsk, M. A. (2004). Visual attention as a multilevel selection process. *Cognitive, Affective, & Behavioral Neuroscience*, 4(4), 483-500.
- Laeng, B., & Teodorescu, D. S. (2002). Eye scanpaths during visual imagery reenact those of perception of the same visual scene. *Cognitive Science*, 26(2), 207-231.
- McGowan, J. W., Kowler, E., Sharma, A., & Chubb, C. (1998). Saccadic localization of random dot targets. *Vision research*, 38(6), 895-909.
- Miller, G. A., Galanter, E., & Pribram, K. H. (1960). *Plans and the Structure of Behavior*.
- Neisser, U. (2014). *Cognitive psychology: Classic edition*. Psychology Press.
- O'regan, J. K. (1992). Solving the "real" mysteries of visual perception: the world as an outside memory. *Canadian Journal of Psychology/Revue canadienne de psychologie*, 46(3), 461.
- Palmer, J. (1988). Very short-term visual memory for size and shape. *Perception & Psychophysics*, 43(3), 278-286.
- Palmer, J. (1990). Attentional limits on the perception and memory of visual information. *Journal of Experimental Psychology: Human perception and performance*, 16(2), 332.

- Posner, M. I. (1980). Orienting of attention. *Quarterly journal of experimental psychology*, 32(1), 3-25.
- Richardson, D. C., & Spivey, M. J. (2000). Representation, space and Hollywood Squares: Looking at things that aren't there anymore. *Cognition*, 76(3), 269-295.
- Schmidt, B. K., Vogel, E. K., Woodman, G. F., & Luck, S. J. (2002). Voluntary and automatic attentional control of visual working memory. *Perception & psychophysics*, 64(5), 754-763.
- Scholz, A., Mehlhorn, K., Bocklisch, F., & Krems, J. (2011). Looking at nothing diminishes with practice. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 33, No. 33).
- Scholz, A., Mehlhorn, K., & Krems, J. F. (2016). Listen up, eye movements play a role in verbal memory retrieval. *Psychological research*, 80(1), 149-158.
- Scholz, A., Krems, J. F., & Jahn, G. (2017). Watching diagnoses develop: Eye movements reveal symptom processing during diagnostic reasoning. *Psychonomic bulletin & review*, 24(5), 1398-1412.
- Scholz, A., Klichowicz, A., & Krems, J. F. (2018). Covert shifts of attention can account for the functional role of "eye movements to nothing". *Memory & cognition*, 46(2), 230-243.
- Smith, E. E., Jonides, J., Koeppe, R. A., Awh, E., Schumacher, E. H., & Minoshima, S. (1995). Spatial versus object working memory: PET investigations. *Journal of cognitive neuroscience*, 7(3), 337-356.
- Sperling, G. (1960). The information available in brief visual presentations. *Psychological monographs: General and applied*, 74(11), 1
- Spivey, M. J., & Geng, J. J. (2001). Oculomotor mechanisms activated by imagery and memory: Eye movements to absent objects. *Psychological research*, 65(4), 235-241.
- Theeuwes, J., Belopolsky, A., & Olivers, C. N. (2009). Interactions between working memory, attention and eye movements. *Acta psychologica*, 132(2), 106-114.
- Theeuwes, J., Kramer, A. F., & Irwin, D. E. (2011). Attention on our mind: The role of spatial attention in visual working memory. *Acta psychologica*, 137(2), 248-251.

Vogel, E. K., Luck, S. J., & Shapiro, K. L. (1998). Electrophysiological evidence for a postperceptual locus of suppression during the attentional blink. *Journal of Experimental Psychology: Human Perception and Performance*, 24(6), 1656.